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RESEARCH OVERVIEW

I work on the stochastic modeling of cloud microphysical processes in turbulence and on the causal inference needed to decide, from data, which of those processes drives which. The method is the same in every project: reduce the process to the smallest model that keeps the mechanism, push the analysis as far as a closed form allows, verify the numerics against those limits and against convergence in time step, ensemble size and seed, then ask what the model says about measurements. The models, the analysis, the numerics and the software are my own work.

The physics thread. Supersaturation in turbulent clouds fluctuates over a wide range of time scales, so droplets with different supersaturation histories grow at different rates and the cloud droplet size distribution (CDS) broadens beyond mean-field parcel theory. I represent condensational growth, the fluctuating saturation ratio and entrainment-mixing as coupled stochastic processes in Langevin and Fokker-Planck form, with liquid water content and droplet number concentration conserved, and study how the correlation time and intensity of supersaturation fluctuations set spectral broadening and relative dispersion.

The inference thread treats those same models as testbeds. Aerosol-cloud-turbulence interactions are high-dimensional and nonlinear, the couplings run in both directions, and the drivers that matter are often unmeasured, so correlation cannot say which process forces which. I embed measures of directed influence (Liang-Kleeman information flow, transfer entropy, conditional mutual information) in stochastic condensation models whose causal structure is known analytically, and use them as ground truth for causal discovery that tolerates feedback and unobserved drivers. The implementation takes any spatiotemporal record, and I stress-test it outside the atmosphere on financial series, where the sampling interval is fixed and no ground truth exists. The goal is a process-level, memory-aware account of droplet spectral evolution that can inform microphysics parameterizations.

EDUCATION

Ph.D., Applied Mathematics and Statistics

2022 – May 2027 (expected)

Stony Brook University, Stony Brook, NY

Dissertation: *Uncovering Causal Mechanisms of Aerosol-Cloud-Turbulence Interactions Using Physics- and Data-Driven Methods*

Advisors: Xiaolin Li (Stony Brook University) and Yangang Liu (Brookhaven National Laboratory)

Advanced to candidacy, Fall 2025. GPA: 3.78 / 4.0

M.S., Finance, University of Rochester, Rochester, NY. GPA: 3.81 / 4.0

Aug 2021 – Aug 2022

Certificate in Quantitative Finance (CQF), CQF Institute

RESEARCH EXPERIENCE

Graduate Research Assistant, Brookhaven National Laboratory

Jan 2024 – Jan 2026

Environmental Science and Technologies Department, Upton, NY

Sole Ph.D. student on a joint program of Brookhaven and Applied Mathematics and Statistics at Stony Brook on stochastic condensation and causal analysis of aerosol-cloud-turbulence interactions, responsible for direction, theory, code and manuscripts. Supported by the DOE Atmospheric System Research (ASR) Program and BNL Laboratory Directed Research and Development (LDRD-25-008). The four themes below were carried out there.

Memory of supersaturation fluctuations and spectral broadening

- Generalized the McGraw and Liu (2006) Brownian drift-diffusion model, valid only in the strongly damped limit, by representing the saturation ratio as an Ornstein-Uhlenbeck process coupled to the condensational growth equation. The decorrelation rate spans the full memory continuum, and the McGraw-Liu model is recovered as it goes to infinity [1].
- Lagrangian Monte Carlo conserving liquid water content and droplet number concentration, with time-step and ensemble-size convergence tests, shows relative dispersion increasing with both the correlation time and the standard deviation of supersaturation fluctuations, the white-noise model underestimating broadening at long correlation times, and the dependence collapsing on the ratio of fluctuation intensity to decorrelation rate [1].

Information flow between supersaturation and droplet growth

- Incorporated the Liang-Kleeman information-flow formalism into the coupled model, derived the supersaturation-to-droplet information flow in closed form with its white-noise and frozen-noise lim-

its, and estimated it from ensemble covariances. The estimate matches theory, the reverse flow vanishes, and the influence of supersaturation on droplet size weakens as the CDSB broadens, which correlation cannot state [1].

Maximum entropy systems theory and Langevin stochastic condensation

- Maximum entropy systems theory predicts a Weibull CDSB whose order follows from the conserved quantity but has no dynamics; Langevin stochastic condensation has dynamics, but its spectral order is fixed by the variable in which the noise is additive. Showed that their disagreement under mass conservation reflects the choice of size variable, and generalized stochastic condensation to an arbitrary conserved moment order under the equal-shares hypothesis, imposed through the growth law, the saturation ratio a droplet experiences, or both [2].
- Showed that the conserved order sets the spectral shape while the normalizing moment sets only the scale, and that the residual underestimate of relative dispersion traces to the small-size boundary, where no rule can conserve probability, leave the saturation ratio unchanged, and avoid accumulation at zero size at once. The three formulations share a size spectrum but differ in the joint statistics of droplet size and local saturation ratio, so observations can discriminate among them [2].

Entrainment-mixing in a stochastic condensation framework

- Built a three-state stochastic condensation and entrainment-mixing model: an adiabatic steady state, an entrainment event by unbiased thinning, and extreme-inhomogeneous (complete evaporation) and homogeneous (uniform shrinkage) pathways to the same target liquid water content. The pathways produce distinct spectral evolutions and distinct trajectories of the homogeneous mixing degree measures of Lu et al. (2013) [3].
- Formulated a general mixing operator spanning the continuum between the two limits, with controlled fluctuations of droplet number and liquid water content, to probe how entrainment-induced variability reshapes the CDSB and to constrain mixing parameterizations [3].

Doctoral Research, Stony Brook University

Aug 2023 – Present

Department of Applied Mathematics and Statistics, Stony Brook, NY

The dissertation places the studies above inside one question: when a population of particles is driven by a turbulent environment that is observed only in part, what can be decided about cause from the data that exist? Its methodological half builds the inference machinery and the software, and trusts a method on observations only once it recovers an answer that theory already fixes.

Causal discovery with feedback and unobserved drivers

- Developing a causal discovery framework for high-dimensional, nonlinear spatiotemporal geophysical data that assumes neither acyclicity nor causal sufficiency: partial ancestral graphs over a lag window under sigma-separation, nearest-neighbor conditional mutual information tests, orientation from time precedence and nonstationarity, a surrogate-calibrated diagnostic separating nonlinearity from nonstationarity, and a path-space relative entropy rate (closed form by Girsanov's theorem) as a time-resolved measure of causal strength. Validation uses stochastic condensation systems whose directed influences are known [4].
- A sampling diagnostic that reports when the observing interval is too coarse for direction to be identified, so the method declines to orient an edge rather than report a spurious one. Exercised on geophysical and on financial records, the latter because their sampling interval cannot be refined at will.

RESEARCH PLAN

The direction is a single stochastic framework in which supersaturation fluctuations, activation, entrainment-mixing and the resulting spectra all appear, and in which every causal claim made about it is first tested on a case whose answer is known.

- *Two-way coupling and vapor-liquid feedback.* Let condensation deplete supersaturation, which opens causal attribution of vapor-liquid feedback and gives analytically controlled feedback benchmarks.
- *Activation and deactivation.* Add curvature and solute effects at the small-size end in place of idealized boundary closures, and quantify how closure and conservation constraints shape the CDSB.
- *Entrainment-mixing and supersaturation variability.* Couple the general mixing operator to finite-memory fluctuations, linking mixing regime, droplet number and liquid water variability, and broadening.
- *Nonlinear causal attribution.* Move past linear-Gaussian information-flow estimators with transfer entropy and conditional mutual information inside causal machine learning, for non-Gaussian regimes.
- *Observational constraints.* Test memory-dependent broadening, information transfer and the predicted size and supersaturation joint statistics against cloud chamber and field data, including long-term ARM records.

PUBLICATIONS

Journal articles

- [1] **Shu, Y.**, Liu, Y., Zhang, T., McGraw, R., and Li, X., 2026: Stochastic condensation with information flow for causality analysis. *Geophysical Research Letters*, **53**(17), e2026GL123056. doi:10.1029/2026GL123056. Open access.

Manuscripts in preparation

- [2] **Shu, Y.**, Liu, Y., and Li, X.: A Langevin-type stochastic condensation model to explain the size distribution based on maximum entropy principle. For *Physical Review E*.
- [3] **Shu, Y.**, Liu, Y., and Li, X.: A new stochastic condensation model for investigating turbulent entrainment-mixing processes. For *Journal of the Atmospheric Sciences*.
- [4] **Shu, Y.**: Entropy-based causal discovery for spatiotemporal dynamical systems with feedback and unmeasured drivers. Methodology working paper.

Research software

- `stochastic_condensation_info_flow` (v1.2), code for [1]. MIT License. doi:10.5281/zenodo.22089240; GitHub.
- `ecausal`, the implementation of [4]: an installable Python package and command-line tool that takes any multivariate or spatiotemporal record and returns a partial ancestral graph whose edges carry orientation evidence, a linearity verdict and a strength estimate, with nearest-neighbor and copula independence tests, surrogate calibration, bootstrap intervals, FDR control, SLURM profiles, and a test suite pinning every estimator to a closed form or an exact oracle.

CONFERENCE ABSTRACTS AND PRESENTATIONS

- Liu, Y., **Shu, Y.**, Zhang, T., and McGraw, R., 2026: A new generalized stochastic model and impacts of microphysical fluctuations. *106th AMS Annual Meeting*, January 2026 (presented by Y. Liu).
- **Shu, Y.**, Liu, Y., Zhang, T., and McGraw, R. L., 2025: Stochastic condensation/evaporation with different physical constraints and information flow for causal discovery. Poster, *AGU Fall Meeting 2025*, A21B-2034.
- Liu, Y., Gao, S., Lu, C., and **Shu, Y.**, 2025: Relationship between turbulent entrainment-mixing and droplet clustering. *AGU Fall Meeting 2025*, A21B-203.
- Liu, Y., **Shu, Y.**, Zhang, T., and McGraw, R., 2025: Stochastic condensation with information flow for causality analysis. Poster, *DOE Atmospheric System Research (ASR) Program Meeting*, 2025.

TEACHING

Graduate Teaching Assistant, AMS 595 Fundamentals of Computing, Stony Brook University Fall 2026

TECHNICAL SKILLS

Theory: Itô calculus, Langevin and Fokker-Planck equations, Ornstein-Uhlenbeck processes, maximum entropy, moment and boundary closures, asymptotic limits (strongly damped, frozen noise).

Numerics: Explicit and implicit SDE integrators, Lagrangian Monte Carlo under conservation constraints, convergence studies in time step, ensemble size and seed, checks against closed-form steady states, factorial sweeps.

Inference: Liang-Kleeman information flow, transfer entropy, nearest-neighbor conditional mutual information, partial ancestral graphs (FCI, LPCMCI), surrogate tests, bootstrap intervals, false-discovery-rate control.

Computing: Python (NumPy, SciPy, pandas, statsmodels, Numba, PyArrow), MATLAB, C++, R; SLURM clusters and parallel ensembles; Git, pytest, packaging; LaTeX and journal-standard figures.